Notes: ECMA308 (W26) Theory of Auctions

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1 Preliminaries

1.1 Statistical Facts

Proposition 1.1. *Consider k values drawn independently from f. Let $G_{1,k}$ and $G_{2,k}$, respectively, denote the cdf of the highest and second highest among the k values. We have*

$$G_{1,k}(x) = F(x)^k,$$
$$G_{2,k}(x) = F(x)^k + kF(x)^{k-1}(1 - F(x)).$$

The corresponding densities are

$$g_{1,k}(x) = kF(x)^{k-1}f(x),$$
$$g_{2,k}(x) = k(k-1)F(x)^{k-2}(1 - F(x))f(x).$$

These formulas are intuitive if we interpret $f(x)$ as the probability that x is drawn. For instance, $f(x)$ is the probability that the second highest value is x, $1 - F(x)$ is the probability that one of the other $k - 1$ values exceeds x, and $F(x)^{k-2}$ is the probability that the remaining $k - 2$ values are less than x.

1.2 The Four Standard Auctions

(i) First-price sealed-bid: each bidder submits a sealed bid to the seller. The high bidder wins and pays her bid. Note that we did not specify what happens in the event of a tie (they happen with probability 0).

(ii) Second-price sealed-bid: each bidder submits a sealed bid to the seller. The high bidder wins and pays the second highest bid.

(iii) Dutch: The public price starts very high (no takers) and decreases until a bidder raises their hand, at which point that bidder wins and pays the current price.

(iv) English: The seller starts with a price of zero and increases it. Each bidder signals when he wishes to drop out. Once out, cannot return. The last remaining bidder wins at the current price.

2 The Independent Private Values Model

- One unit of an indivisible good.
- N bidders.
- The ith bidder's private value is $V_i \sim f_i$. Values are drawn independently across bidders, and distributions f_i are common knowledge. Choosing units appropriately, assume $V_i \in [0, 1]$ for each i.
- Bidder i has utility $v_i - p$ if she wins the good and pays p, and $-p$ if she does not win and pays p. Bidders maximize expected utility.
- Seller maximizes expected revenue.

2.1 The First-Price Auction

Assumption 2.1.

- Common distribution of values $f_i = f$ for each i.
- Symmetric equilibrium bidding function $\hat{b} : [0, 1] \to [0, \infty)$ mapping values to bids.

For bidder i, let $u_i(r, v)$ denote her expected utility when her value is v and she bids $\hat{b}(r)$, given that all other bidders bid according to $\hat{b}(\cdot)$. We assume now and will later verify that $\hat{b}(v)$ is strictly increasing in v.

We have

$$u_i(r, v) = \mathbb{P}[\text{win}|\text{bid } \hat{b}(r)] \cdot (v - \hat{b}(r)) = F(r)^{N-1}(v - \hat{b}(r)),$$

where the second equality comes from noting that i wins if and only if $\hat{b}(r) > \hat{b}(v_j)$, $\forall j \neq i$, or equivalently $r > v_j$, $\forall j \neq i$.

The symmetric equilibrium imposes that $u_i(r, v)$ is maximized at $r = v$ for all v, or

$$F^{N-1}(r)\left(v - \hat{b}(r)\right) \leq F^{N-1}(v)\left(v - \hat{b}(v)\right), \quad \forall r.$$

Thus, taking the first order condition with respect to r and evaluating at $r = v$ gives

$$(N-1)f(v)F^{N-2}(v)(v - \hat{b}(v)) - F^{N-1}(v)\hat{b}'(v) = 0.$$

Rearranging gives the differential equation

$$(N-1)F^{N-2}(v)f(v)\hat{b}(v) + F^{N-1}(v)\hat{b}'(v) = (N-1)F^{N-2}(v)f(v)v, \quad \forall v \in [0, 1].$$

Noting that the left hand side is the derivative of $F^{N-1}(v)\hat{b}(v)$, we have

$$\frac{\mathrm{d}}{\mathrm{d}v}\left(F^{N-1}(v)\hat{b}(v)\right) = (N-1)F^{N-2}(v)f(v)v, \quad \forall v \in [0, 1].$$

One integrates to get

$$F^{N-1}(v)\hat{b}(v) = \int_0^v (N-1)F^{N-2}(t)f(t)t \, \mathrm{d}t + 0,$$

where the constant 0 comes from $F(0) = 0$. Thus,

$$\hat{b}(v) = \frac{N-1}{F^{N-1}(v)} \int_0^v tF^{N-2}(t)f(t) \, \mathrm{d}t.$$

Exercise 2.2. Compute $\hat{b}$ when f is uniform on $[0, 1]$.

Exercise 2.3. Verify that $\hat{b}$ is strictly increasing and $\hat{b}(v) < v$.

Note that at this point we have only shown that if there exists a symmetric equilibrium with strictly increasing bidding function, it must be $\hat{b}$. To show the existence of such an equilibrium, we need to verify that $\hat{b}$ is indeed an equilibrium bidding function.

Exercise 2.4. Show that $\hat{b}$ is indeed an equilibrium bidding function.

It turns out that this is actually the only equilibrium for this problem (mixed or pure, increasing or otherwise, symmetric or asymmetric).

2.2 Dutch Auction

The Dutch auction is strategically equivalent to the first-price auction: each strategy can be identified by the point at which the bidder raises their hand (since no information is gained before anyone raises their hand).

The only relevant factor is the price at which the bidder raises their hand. Call this price p_i for bidder i. The winner is the bidder with the highest p_i, who pays p_i. This is equivalent to FPA. So, the equilibrium bidding strategy is to raise your hand at price $p = \hat{b}(v)$ when your value is v.

2.3 Second-Price Auction

Suppose each bidder has her value drawn from f_i (here, we allow asymmetric bidders).

In this case bidding your value is a weakly dominant strategy, i.e., the resulting payoff is weakly larger than any other bidding strategy regardless of what the other bidders do. Fix some bidder i and let B denote the highest bid among the other bidders (unknown to i). Suppose i's value is v_i. Player i wants to win when $v_i > B$ and lose when $v_i < B$. Bidding exactly v_i achieves this regardless of what B is.

2.4 English Auction

The English auction is strategically equivalent to the second-price auction.

2.5 Revenue Equivalence

Recall that SPA and EA raise the same expected revenue, and FPA and DA raise the same expected revenue. It is thus enough to compare SPA with FPA.

Recall that in a FPA, bidders bid according to

$$\hat{b} = \frac{N-1}{F^{N-1}(v)} \int_0^v t F^{N-2}(t) f(t) \, \mathrm{d}t,$$

while in a SPA, bidders bid their values. Let R_{FPA} and R_{SPA} respectively denote the expected revenue of the FPA and SPA. Let $g_{1,N}(v) = NF^{N-1}(v)f(v)$ denote the pdf of the highest value among N values drawn iid from f. Then

$$\begin{aligned}
R_{\text{FPA}} &= \int_0^1 \hat{b}(v) g_{1,N}(v) \, \mathrm{d}v \\
&= \int_0^1 \left[\frac{N-1}{F^{N-1}(v)} \int_0^v x f(x) F^{N-2}(x) \, \mathrm{d}x \right] g_{1,N}(v) \, \mathrm{d}v \\
&= N(N-1) \int_0^1 \int_0^v x f(x) F^{N-2}(x) f(v) \, \mathrm{d}x \, \mathrm{d}v \\
&= N(N-1) \int_0^1 x f(x) F^{N-2}(x) \left[\int_x^1 f(v) \, \mathrm{d}v \right] \mathrm{d}x \\
&= N(N-1) \int_0^1 x f(x) F^{N-2}(x) \left[1 - F(x) \right] \mathrm{d}x.
\end{aligned}$$

In line 4 we reparametrized the areas of integration $0 \le x \le v \le 1$.

Let $g_{2,N}(v) = N(N-1)F^{N-2}(v)f(v)[1-F(v)]$ denote the pdf of the second highest value among N values drawn iid from f. Then

$$\begin{aligned}
R_{\text{SPA}} &= \int_0^1 v g_{2,N}(v) \, \mathrm{d}v \\
&= N(N-1) \int_0^1 v F^{N-2}(v) f(v) [1 - F(v)] \, \mathrm{d}v.
\end{aligned}$$

Thus when $f_i = f$ for each i, all 4 standard auctions yield the same expected revenue.

2.6 Interpretation of the Equilibrium Bidding Function

Proposition 2.5.

$$\hat{b}(v) = \mathbb{E}\left[\textit{highest value of others} \mid \textit{my value is } v \textit{ and is highest} \right].$$

Proof. The pdf of the highest value of the others is

$$g_{1,N-1}(x) = (N-1) f(x) F^{N-2}(x).$$

Thus the conditional density given $x \leq v$ is

$$\frac{g_{1,N-1}(x)}{F^{N-1}(v)}.$$

We can now write the right hand side of the desired identity as

$$\begin{aligned}
&\mathbb{E}[\text{highest value of others} \mid \text{all others values} \leq v, \text{ my value is } v] \\
&= \mathbb{E}[\text{highest value of others} \mid \text{all others values} \leq v] \\
&= \frac{1}{F^{N-1}(v)} \int_0^v x(N-1) f(x) F^{N-2}(x) \, \mathrm{d}x \\
&= \hat{b}(v).
\end{aligned}$$

□

This gives another proof for the following result (which can be obtained through direct computation).

Corollary 2.6.

(i) $\hat{b}(v) < v$ *if* $v \neq 0$.

(ii) Note that

$$\begin{aligned}
\hat{b}(v) &= \mathbb{E}[\textit{2nd highest value} \mid \textit{my value is } v \textit{ and highest}] \\
&= \mathbb{E}[\textit{2nd highest value} \mid \textit{highest value is } v] \\
&= R_{SPA}|_{\textit{highest value is } v},
\end{aligned}$$

where the second line is by symmetry of the bidders. But since by definition of $\hat{b}$, $R_{FPA}|_{\textit{highest value is } v} = \hat{b}(v)$, *we have* $R_{FPA} = R_{SPA}$ *for each realization of the highest value, and thus also in expectation.*

3 Direct Selling Mechanisms

Definition 3.1. A **direct selling mechanism** (DSM) is a collection of N **probability assignment functions** $p_i(v_1, \ldots, v_N)$ and N **cost functions** $c_i(v_1, \ldots, v_N)$ such that for each bidder i and every vector of values $v_1, \ldots, v_N$, $p_i(1, \ldots, v_N) \in [0, 1]$ denotes the probability that bidder i receives the good for sale and $c_i(v_1, \ldots, v_N)$ denotes bidder i's cost (payment to the seller), where $\sum_i p_i \leq 1$.

A DSM works as follows:

(i) Bidders report (privately) their values $r_1, \ldots, r_N$ (possibly false).

(ii) The seller allocates the good according to $p_1(r_1, \ldots, r_N), \ldots, p_N(r_1, \ldots, r_N)$ and the bidders pay their costs $c_1(r_1, \ldots, r_N), \ldots, c_N(r_1, \ldots, r_N)$.

We will consider a subset of DSMs that are **incentive compatible**.

Fix a DSM (p, c). For each bidder i and for each $r_i, v_i \in [0, 1]$, write

$$\begin{aligned} u(r_i, v_i) &:= \int_0^1 \cdots \int_0^1 [p_i(r_i, v_{-i}) \cdot v_i - c_i(r_i, v_{-i})]\, f_1(v_1) \ldots f_{i-1}(v_{i-1}) f_{i+1}(v_{i+1}) \ldots f_N(v_N)\, \mathrm{d}v_1 \ldots \mathrm{d}v_N \\ &= \int [p_i(r_i, v_{-i}) \cdot v_i - c_i(r_i, v_{-i})] f_{-i}(v_{-i})\, \mathrm{d}v_{-i} \end{aligned}$$

for the expected utility of bidder i when she reports r_i and her true value is v_i, given that all other bidders report truthfully.

We can simplify this expression by defining

$$\begin{aligned} \overline{p}_i(r_i) &:= \int p_i(r_i, v_{-i}) f_{-i}(v_{-i})\, \mathrm{d}v_{-i}, \\ \overline{c}_i(r_i) &:= \int c_i(r_i, v_{-i}) f_{-i}(v_{-i})\, \mathrm{d}v_{-i} \end{aligned}$$

to be the expected probability of winning and expected cost when bidder i reports r_i and all other bidders report truthfully. Then,

$$u_i(r_i, v_i) = \overline{p}_i(r_i) \cdot v_i - \overline{c}_i(r_i).$$

Definition 3.2 (Incentive Compatibility)**.** An **incentive-compatible direct selling mechanism** (ICDSM) is a DSM such that each bidder i can do not better than to report truthfully given that all other bidders always report truthfully, i.e., for every i and every $v_i, r_i \in [0, 1]$,

$$v_i = \operatorname*{arg\,max}_{r_i \in [0,1]} u_i(r_i, v_i) = \operatorname*{arg\,max}_{r_i \in [0,1]} \left[\overline{p}_i(r_i) \cdot v_i - \overline{c}_i(r_i)\right].$$

Proposition 3.3. *Each of the four standard auctions can be implemented with an ICDSM.*

Proof. We start with second-price auction. Ignoring ties, define

$$p_i^{\mathrm{SPA}}(v_1, \ldots, v_N) := \begin{cases} 1, & v_i = \max_j v_j, \\ 0, & v_i < \max_j v_j, \end{cases}, \quad c_i^{\mathrm{SPA}}(v_1, \ldots, v_N) := \begin{cases} v_{(2)}, & v_i = \max_j v_j, \\ 0, & v_i < \max_j v_j \end{cases}.$$

Since bidding your value is a weakly dominant strategy in SPA, it is in particular a best response when all other bidders bid truthfully. Thus, $(p^{\mathrm{SPA}}, c^{\mathrm{SPA}})$ is an ICDSM implementation of SPA.

Now consider the English auction. Think really hard and put $(p_i^{\mathrm{EA}}, c_i^{\mathrm{EA}}) := (p_i^{\mathrm{SPA}}, c_i^{\mathrm{SPA}})$. Mr. Auctioneer can tell the bidders that he will implement their optimal strategies if they tell him their values truthfully.

For the first-price auction, assume $f_i = f$ for all i. Mr. Auctioneer can tell the bidders the same thing he did as in the English auction with the $\hat{b}$ we derived. Getting scarily close to a proof by tautology, it appear that we can have an auctioneer tell bidders that he will run a certain auction to show that the auction is compatible with a DSM.[1] If all bidders other than i report truthfully, then they all bid according to $\hat{b}$. Thus by our previous derivation player i's dominant strategy is to bid according to $\hat{b}$; i.e., report v_i. Formally, define

$$p_i^{\mathrm{FPA}}(v_1, \ldots, v_N) := \begin{cases} 1, & v_i = v_{(1)}, \\ 0, & v_i < v_{(1)}, \end{cases}, \quad c_i^{\mathrm{FPA}}(v_1, \ldots, v_N) := \begin{cases} \hat{b}(v_i), & v_i = v_{(1)}, \\ 0, & v_i < v_{(1)} \end{cases}.$$

Similarly, the Dutch auction can be implemented as an ICDSM by defining $(p_i^{\mathrm{DA}}, c_i^{\mathrm{DA}}) := (p_i^{\mathrm{FPA}}, c_i^{\mathrm{FPA}})$. □

We have shown the following:

Theorem 3.4 (Special Revelation Principle)**.** *For each of the four standard auctions and their equilibria, we can define a DSM (p, c) in which*

- *Truth-telling is an equilibrium; i.e., (p, c) is an ICDSM.*
- *In the truth-telling equilibrium of (p, c), for all i and $(v_1, \ldots, v_N)$, i's winning probability p_i and cost c_i are exactly the same as in the equilibrium of the given standard auction.*

Hence, all bidders and the seller are indifferent between the auction and the DSM.

We have not shown the following, which justifies restricting attention to ICDSM when characterizing *equilibrium outcomes* (allocations and payments) that are implementable by some selling mechanism.

Theorem 3.5 (Revelation Principle)**.** *For any selling mechanism and any of its equilibria, we can define a DSM (p, c) in which*

- *Truth-telling is an equilibrium; i.e., (p, c) is an ICDSM.*
- *In the truth-telling equilibrium of (p, c), for all i and $(v_1, \ldots, v_N)$, i's winning probability p_i and cost c_i are exactly the same as in the equilibrium of the given standard auction.*

Hence, all bidders and the seller are indifferent between the selling mechanism and the DSM.

[1] This suffices as a proof for any auction with an equilibrium being implementable as an ICDSM.

Proposition 3.6 (Characterization of Incentive Compatibility)**.** *A DSM* (p, c) *is incentive-compatible if for each i,*

- $\overline{p}_i$ *is nondecreasing in* v_i *on* $[0, 1]$*, and*
- $\overline{c}_i(v_i) = \overline{c}_i(0) + \overline{p}_i(v_i)v_i - \int_0^{v_i} \overline{p}_i(x)\,\mathrm{d}x$.

Proof. We assume as much differentiability as needed.[2]

$\Rightarrow$: Recall $u_i(r_i, v_i) = \overline{p}_i(r_i)v_i - \overline{c}_i(r_i)$. Incentive compatibility implies the FOC

$$0 = \partial_{r_i} u_i(r_i, v_i)|_{r_i=v_i} = \overline{p}_i'(v_i)v_i - \overline{c}_i'(v_i), \quad \forall v_i.$$

Thus, applying integration by parts,

$$\overline{c}_i(v_i) = \overline{c}_i(0) + \int_0^{v_i} \overline{p}_i'(x)x\,\mathrm{d}x = \overline{c}_i(0) + \overline{p}_i(v_i)v_i - \int_0^{v_i} \overline{p}_i(x)\,\mathrm{d}x.$$

Next, we show that $\overline{p}_i$ is nondecreasing. The SOC for maximizing $u_i(r_i, v_i)$ at $r_i = v_i$ gives

$$\left.\frac{\mathrm{d}^2 u_i(r_i, v_i)}{\mathrm{d}r^2}\right|_{r_i=v_i} = \overline{p}_i''(r_i)v_i - \overline{c}_i''(r_i)\big|_{r_i=v_i} = \overline{p}_i''(v_i)v_i - \overline{c}_i''(v_i) \le 0, \quad \forall v_i.$$

Now recall that $\overline{p}_i'(v_i)v_i = \overline{c}_i'(v_i)$ for each v_i. Differentiating gives $\overline{p}_i''(v_i)v_i + \overline{p}_i'(v_i) = \overline{c}_i''(v_i)$. Substituting into the SOC inequality gives

$$\overline{p}_i''(v_i)v_i - \left[\overline{p}_i''(v_i)v_i + \overline{p}_i'(v_i)\right] = -\overline{p}_i'(v_i) \le 0, \quad \forall v_i,$$

$\Leftarrow$: Note that the second condition implies $\overline{p}_i'(v_i)v_i = \overline{c}_i'(v_i)$ for each v_i. Now,

$$\frac{\mathrm{d}u(r_i, v_i)}{\mathrm{d}r_i} = \overline{p}_i'(r_i)v_i - \overline{c}_i'(r_i) = \overline{p}_i'(r_i)\,(v_i - r_i)\,.$$

Since $\overline{p}_i' \ge 0$ (by being nondecreasing), we have that $\mathrm{d}u(r_i, v_i)/\mathrm{d}r_i \le 0$ when $r_i \ge v_i$ and $\mathrm{d}u(r_i, v_i)/\mathrm{d}r_i \ge 0$ when $r_i \le v_i$. Thus, $u(r_i, v_i)$ is maximized at $r_i = v_i$. □

Note that the restriction imposed by incentive compatibility left us with only the flexibility to choose $\overline{p}$ and $\overline{c}(0)$. The following result then should not be surprising:

Theorem 3.7 (Revenue Equivalence)**.** *If two ICDSMs have the same probability assignment functions and every bidder with value zero is indifferent between the two mechanisms, then they yield the same expected revenue to the seller.*

Proof. Let $(\overline{p}, \overline{c}^0)$ and $(\overline{p}, \overline{c}^1)$ be ICDSMs. Let R^k be the revenue from the mechanism $k = 0, 1$. We have

$$\begin{aligned} R^k &= \sum_{i=1}^N \int_0^1 \overline{c}_i^k(v_i) f_i(v_i)\,\mathrm{d}v_i \\ &= \sum_{i=1}^N \int_0^{v_i} \left[\overline{c}_i^k(0) + \overline{p}_i(v_i)v_i - \int_0^1 \overline{p}_i(x)\,\mathrm{d}x\right] f_i(v_i)\,\mathrm{d}v_i. \end{aligned}$$

[2] One can prove this result without these assumptions.

Also, since i is indifferent between (p, c^0) and (p, c^1) when $v_i = 0$, we have

$$\overline{p}_i(0) \cdot 0 - \overline{c}_i^0(0) = \overline{p}_i(0) \cdot 0 - \overline{c}_i^1(0) \implies \overline{c}_i^0(0) = \overline{c}_i^1(0).$$

Thus R^k does not depend on k. □

Example 3.8. We can use the Revenue Equivalence Theorem to solve for $\hat{b}(v)$ for FPA. Suppose $\hat{b}(v)$ is an (symmetric) equilibrium of a FPA. If i's value is v, her expected cost is

$$\overline{c}_i(v) = \hat{b}(v) \cdot \mathbb{P}[\text{win}|\text{value } v] = \hat{b}(v)F^{N-1}(v).$$

Her probability of winning is $\overline{p}_i(v) = F^{N-1}(v)$. By incentive compatibility,

$$F^{N-1}(v)\hat{b}(v) = 0 + F^{N-1}(v)v - \int_0^v F^{N-1}(x)\,\mathrm{d}x.$$

Thus,

$$\hat{b}(v) = v - \int_0^v \left(\frac{F(x)}{F(v)}\right)^{N-1} \mathrm{d}x.$$

4 Revenue Maximizing DSM

Definition 4.1. A DSM (p, c) is **individually rational** (IR) if for each bidder i and each $v_i \in [0, 1]$,

$$\overline{p}_i(v_i)v_i - \overline{c}_i(v_i) \geq 0.$$

Thus IR DSMs guarantee that each bidder is weakly better off participating in the mechanism than not participating.

Proposition 4.2. *If (p, c) is IC, then (p, c) is IR if and only if $\overline{c}_i(0) \leq 0$ for each i.*

Proof. $\Rightarrow$: IR implies

$$\overline{p}_i(0) \cdot 0 - \overline{c}_i(0) \geq 0$$

for each i.

$\Leftarrow$: Suppose (p, c) is IC and $\overline{c}_i(0) \leq 0$ for each i. From IC we know

$$\overline{p}_i(v_i)v_i - \overline{c}_i(v_i) \geq \overline{p}_i(0)v_i - \overline{c}_i(0) \geq -\overline{c}_i(0) \geq 0.$$

□

We can now find the seller's maximum possible revenue among all selling mechanisms that have equilibria by solving the following problem:

$$\max_{(p,c)} \sum_{i=1}^{N} \int_0^1 \overline{c}_i(v_i) f_i(v_i) \, \mathrm{d}v_i,$$

where (p, c) must satisfy the following IC and IR constraints:

(i) $\overline{p}_i$ nondecreasing in $v_i \in [0, 1]$.

(ii) $\overline{c}_i(v_i) = \overline{c}_i(0) + \overline{p}_i(v_i)v_i - \int_0^{v_i} \overline{p}_i(x) \, \mathrm{d}x$ for each $v_i \in [0, 1]$.

(iii) $\overline{c}_i(0) \leq 0$.

(iv) Implicitly, $\sum p_i(v_1, \ldots, v_N) \leq 1$ and $p_i \geq 0$.

For simplicity, we assume the following:

Assumption 4.3.

$$v_i - \frac{1 - F_i(v_i)}{f_i(v_i)}$$

is strictly increasing in v_i for each i.

This is again not required for the results to hold, but makes the analysis easier.

We first substitute condition (ii) into the objective function to get

$$\begin{aligned} R &= \sum_{i=1}^{N} \int_0^1 \left[\overline{c}_i(0) + \overline{p}_i(v_i)v_i - \int_0^{v_i} \overline{p}_i(x) \, \mathrm{d}x \right] f_i(v_i) \, \mathrm{d}v_i \\ &= \sum_{i=1}^{N} \overline{c}_i(0) + \sum_{i=1}^{N} \left[\int_0^1 \overline{p}_i(v_i)v_i f_i(v_i) \, \mathrm{d}v_i - \int_0^1 \int_0^{v_i} \overline{p}_i(x) f_i(v_i) \, \mathrm{d}x \, \mathrm{d}v_i \right]. \end{aligned}$$

The double integral can be simplified by changing the order of integration:

$$\int_0^1 \int_0^{v_i} \overline{p}_i(x) f_i(v_i)\, \mathrm{d}x\, \mathrm{d}v_i = \int_0^1 \int_x^1 \overline{p}_i(x) f_i(v_i)\, \mathrm{d}v_i\, \mathrm{d}x$$
$$= \int_0^1 \overline{p}_i(x)[1 - F_i(x)]\, \mathrm{d}x.$$

Thus,

$$R = \sum_{i=1}^N \overline{c}_i(0) + \sum_{i=1}^N \left[\int_0^1 \overline{p}_i(v_i) v_i f_i(v_i)\, \mathrm{d}v_i - \int_0^1 \overline{p}_i(v_i)[1 - F_i(v_i)]\, \mathrm{d}v_i \right]$$
$$= \sum_{i=1}^N \overline{c}_i(0) + \sum_{i=1}^N \int_0^1 \left(v_i - \frac{1 - F_i(v_i)}{f(v_i)} \right) \overline{p}_i(v_i) f_i(v_i)\, \mathrm{d}v_i.$$

We now substitute in

$$\overline{p}_i(v_i) = \int p_i(v_i, v_{-i}) f_{-i}(v_{-i})\, \mathrm{d}v_{-i}$$

to get

$$R = \sum_{i=1}^N \overline{c}_i(0) + \sum_{i=1}^N \int_0^1 \int \left(v_i - \frac{1 - F_i(v_i)}{f_i(v_i)} \right) p_i(v_i, v_{-i}) f(v_1) \dots f(v_N)\, \mathrm{d}v_{-i}\, \mathrm{d}v_i$$
$$= \sum_{i=1}^N \overline{c}_i(0) + \int \sum_{i=1}^N \left(v_i - \frac{1 - F_i(v_i)}{f_i(v_i)} \right) p_i(v_1, \dots, v_N) f(v_1) \dots f(v_N)\, \mathrm{d}v_1 \dots\, \mathrm{d}v_N.$$

We will upper bound R and show that this upper bound can be achieved.

One way to obtain upper bounds is to optimize while ignoring some constraints. We will ignore constraint (i), that $\overline{p}_i$ is nondecreasing in v_i for each i. Condition (iii) implies $\sum_i \overline{c}_i(0) \leq 0$. Since we have $\sum_i p_i(v_1, \dots, v_N)$, we can place all the weights of p on the bidder i that maximizes $v_i - (1 - F_i(v_i))/f_i(v_i)$ (or 0, if all are negative). Thus, we have the upper bound

$$R \leq \int \max_i \left[v_i - \frac{1 - F_i(v_i)}{f_i(v_i)} \right]^+ f(v_1) \dots f(v_N)\, \mathrm{d}v_1 \dots\, \mathrm{d}v_N,$$

where for clarity, we write

$$\left[v_i - \frac{1 - F_i(v_i)}{f_i(v_i)} \right]^+ = \max \left\{ v_i - \frac{1 - F_i(v_i)}{f_i(v_i)}, 0 \right\}, \quad \forall i.$$

Now, let's show that this upper bound can be achieved by some ICDSM (p^*, c^*). In order to achieve the upper bound of the weighted average, we must define p_i^* as follows:

$$p_i^*(v_1, \dots, v_N) := \begin{cases} 1, & \left[v_i - \frac{1 - F_i(v_i)}{f_i(v_i)} \right]^+ > \left[v_j - \frac{1 - F_j(v_j)}{f_j(v_j)} \right]^+, \quad \forall j \neq i, \\ 0, & \text{otherwise.} \end{cases}$$

Ties occur with probability 0 (by our strict monotonicity assumption), so we ignore them. Now, in principle, we can use p_i^* to calculate $\overline{p}_i$ and then $\overline{c}_i$ using condition (ii), but this does not give us c^*.

We will define $c_1^*, \dots c_N^*$ so that (ii) holds and we have $\overline{c}_i(0) = 0$. This can be done by defining, say,

$$c_i^*(v_1, \dots, v_N) \coloneqq 0 + p_i^*(v_1, \dots, v_N)v_i - \int_0^{v_i} p_i^*(x, v_{-i})\, \mathrm{d}x.$$

Integrating, we must have conditions (ii) and (iii) (since these conditions holds pointwise, they must hold in expectation). It remains to check (i), that $\overline{p}_i$ is nondecreasing in v_i for each i. Note that p_i^* is nondecreasing in v_i for each fixed v_{-i} by our assumption that $v_i - \frac{1-F_i(v_i)}{f_i(v_i)}$ is strictly increasing in v_i. Thus, the same holds for $\overline{p}_i$.

We have verified that (p^*, c^*) satisfies (i)–(iii) and achieves the upper bound on R. Thus, it is optimal.

Remark 4.4. p^* was essentially unique (recall how it is determined), while there are multiple ways to choose c^*, as long as in expectation they satisfy (ii) and (iii). Compare this to the four standard auctions, which all yield the same expected revenue but have different payment rules. ☕

4.1 Allocation Probabilities, Marginal Revenue

Now, let's interpret $v_i - \frac{1-F_i(v_i)}{f_i(v_i)}$. It turns out that they can be interpreted as "marginal revenues." Recall

$$R = \sum \overline{c}_i(0) + \int \sum_i \left(v_i - \frac{1 - F_i(v_i)}{f_i(v_i)} \right) p_i(v_1, \dots, v_N) f(v_1) \dots f(v_N)\, \mathrm{d}v_1 \dots\, \mathrm{d}v_N.$$

The marginal revenue with respect to increasing bidder i's assignment probability p_i is precisely

$$v_i - \frac{1 - F_i(v_i)}{f_i(v_i)}.$$

Intuition:

Remark 4.5.

- If value is known, the marginal revenue is simply v_i. Agents who assignment probability increases has higher incentive to lie; incentive compatibility imposes a cost.
- Hereafter, we write
$$\mathrm{MR}_i(v_i) \coloneqq v_i - \frac{1 - F_i(v_i)}{f_i(v_i)}$$
as the **marginal revenue** for bidder i, so that
$$p_i^*(v_1, \dots, v_N) = \begin{cases} 1, & \mathrm{MR}_i^+(v_i) > \mathrm{MR}_j^+(v_j), \quad \forall j \neq i, \\ 0, & \text{otherwise.} \end{cases}$$

- Another way to think about MR_i: What price P must I charge bidder i so that I sell to her with probability Q? Well, $P := P_i(Q)$ must solve

$$1 - F_i(P) = Q.$$

Note that this implies

$$-f_i(P_i(Q)) \cdot P_i'(Q) = 1 \implies P_i'(Q) = -\frac{1}{f_i(P_i(Q))}.$$

In this case, the expected revenue from bidder i is $P_i(Q) \cdot Q$ and so

$$\begin{aligned}\mathrm{MR}_i(Q) &= P_i'(Q) \cdot Q + P_i(Q) \\ &= -\frac{Q}{f_i(P_i(Q))} + P_i(Q) = P_i(Q) - \frac{1 - F_i(P_i(Q))}{f_i(P_i(Q))}.\end{aligned}$$

☕

4.2 Cost Functions

Suppose $p_i^*(c_1, \ldots, v_N) = 0$. Then,

$$c_i^*(c_1, \ldots, v_N) = 0 \cdot v_i - \int_0^{v_i} p_i^*(v_i, v_{-i})\, \mathrm{d}x = 0,$$

since p_i^* is nondecreasing in v_i (this follows from MR_i being strictly increasing in v_i).

Proposition 4.6. *If you don't get the good, you don't pay.*

Now suppose $p_i^*(v_1, \ldots, v_N) = 1$. Then

$$\begin{aligned}c_i^*(v_1, \ldots, v_N) &= p_i^*(v_1, \ldots, v_N) v_i - \int_0^{v_i} p_i^*(x, v_{-i})\, \mathrm{d}x \\ &= v_i - \int_0^{v_i} p_i^*(x, v_{-i})\, \mathrm{d}x.\end{aligned}$$

Note that $p_i^*(x, v_{-i}) = 1$ only if $\mathrm{MR}_i > r_i^*$ (MR_i is a strictly increasing function from $\mathrm{MR}_i(0) = -1/f(0)$ to $\mathrm{MR}_i(1) = 1$), where $r_i^* = r_i^*(v_{-i})$ solves $\mathrm{MR}_i(r_i^*) = \max_{j \neq i} \mathrm{MR}_j(v_j)$, or

$$r_i^* - \frac{1 - F_i(r_i^*)}{f_i(r_i^*)} = \max_{j \neq i} \left(v_j - \frac{1 - F_j(v_j)}{f_j(v_j)} \right)^+.$$

Thus,

$$c_i^*(v_1, \ldots, v_N) = v_i - (v_i - r_i^*) = r_i^*.$$

In particular,

Proposition 4.7. *The payment r_i^* depends only on others' values, and is precisely the highest report you could have made without winning. (Compare this to SPA.)*

Proposition 4.8. *Note that the optimal auction is not just incentive-compatible. Truth reporting is a weakly dominant strategy.*

We can describe the optimal auction as follows:

Proposition 4.9. *Given any vector $(v_1, \ldots, v_N)$ of reports, the good is assigned to bidder i whose $MR_i(v_i)$ is the highest and positive (if any). Losers pay nothing and the winner pays the highest value he could have reported without winning.*

4.3 Efficiency

Suppose $N = 2$ with values drawn from f_1, f_2. It is possible that $v_1^0 > v_2^0$ but $0 < \mathrm{MR}_1(v_1^0) < \mathrm{MR}_2(v_2^0)$ (draw a diagram of MR_i against v_i, recalling $\mathrm{MR}_i(0) = -1/f_i(0)$ and $\mathrm{MR}_i(1) = 1$). We see that inefficiency can arise in the optimal auction: bidder 2 wins even though bidder 1 has a higher value. (In this case bidder i pays r_2^*, which is the solution to $\mathrm{MR}_2(r_2^*) = \mathrm{MR}_1(v_1^0)$.)

4.4 The Symmetric Case

Let's get rid of this source of inefficiency by imposing $f_i = f$ for each i so that MR_i are all the same. (Note that we dropped the assumption that $N = 2$.) Since MR_i is strictly increasing, the bidder with the highest value wins so long as their MR is positive. Let ρ^* solve $\mathrm{MR}_i(\rho^*) = 0$, or

$$\rho^* - \frac{1 - F(\rho^*)}{f(\rho^*)} = 0.$$

We have that any bidder with value below ρ^* does not win, and the bidder with the highest value above ρ^* wins and pays the second highest value or ρ^*, whichever is higher (this is the highest value they could have reported without winning). This is precisely a SPA with a **reserve price** of ρ^*. One may think of this as the seller bidding ρ^* in the auction, and a bidder wins only if their value exceeds the seller's bid, paying the second highest bid (including the seller's bid).

Remark 4.10. The FPA with the same reserve price generates the same expected revenue and so it also optimal. Note that the probability assignment functions are the same in both auctions, and bidders with value 0 pay 0 in both auctions. ☕

We will next relax the assumption that values/signals are private and independent.

5 Interdependent-Values Model

The interdependent-values correlated private information model has the following setup:

- 1 unit of indivisible good for sale.
- N bidders, 1 seller.
- Bidder i does not know their value.
- Instead, each bidder i receives a signal s_i about their value v_i. Assume that $s_i, v_i \in [0, \infty)$.
- It is useful to distinguish between random variables and their realizations. To that end, we use upper case letters to denote random variables and lower case letters to denote their realizations.
- We have the following random variables:

  $$V_1, \ldots, V_N, S_1, \ldots, S_N.$$

  Let $f(v_1, \ldots, v_n, s_1, \ldots, s_n)$ be the probability density that $V_i = v_i$ and $S_i = s_i$ for each i.
- We will assume ax ante **symmetry** in each of the auction formats we consider, in the following sense: For each $i \neq j$ and any $(v, s) = (v_1, \ldots, v_N, s_1, \ldots, s_N)$ and $(v', s') = (v'_1, \ldots, v'_N, s'_1, \ldots, s'_N)$, we have

  $$f(v', s') = f(v, s)$$

  whenever v' is obtained from v by switch v_i and v_j and s' is obtained from s by switching s_i and s_j. This implies, for instance, that

  $$\mathbb{E}[V_1|s] = \mathbb{E}[V_1|s']$$

  whenever $(s'_2, \ldots, s'_N)$ is a permutation of $(s_2, \ldots, s_N)$.
- Note that this model is a strict expansion of the independent private value model. In particular, we can recover the independent private value model by assuming that $V_i = S_i$ for each i and that $S_1, \ldots, S_N$ are independent.

We impose also the following structure on signals:

Assumption 5.1 (**Affiliation** (a form of positive correlation)). For each i, each $\overline{v} = (\overline{v}_1, \ldots, \overline{v}_N) \geq (\underline{v}_1, \ldots, \underline{v}_N) = \underline{v}$, and each $\overline{s} \geq \underline{s}$[3],

- $\frac{f(v_i, \overline{v}_{-i}, \overline{s})}{f(v_i, \underline{v}_{-i}, \underline{s})}$ is nondecreasing in v_i on $[0, \infty)$.
- $\frac{f(\overline{v}, s_i, \overline{s}_{-i})}{f(\underline{v}, s_i, \underline{s}_{-i})}$ is nondecreasing in s_i on $[0, \infty)$ and strictly increasing if $\overline{v}_i > \underline{v}_i$.

[3]Here, $x \geq y$ is coordinate wise.

This of this as a monotone likelihood ratio property.

Consequences of symmetry and affiliation:

Proposition 5.2. *Assuming symmetry*[4] *and affiliation, we have the following:*

- $\mathbb{E}[v_1|S_1 = s_1, \ldots, S_N = s_n]$ *is nondecreasing in* s_j *for* $j = 2, \ldots, N$ *and strictly increasing in* s_1.
- $\mathbb{E}[\varphi(S_1, \ldots, S_N)|a_1 \leq S_1 \leq b_1, \ldots, a_N \leq S_N \leq b_N]$ *is nondecreasing in* $a_1, b_1, \ldots, a_N, b_N$ *whenever* φ *is nondecreasing in its arguments.*

Given the symmetry assumed, we will make the following

Assumption 5.3. All bidder use the same equilibrium strategy. We will also assume that bidding is more aggressive when signals are higher. So, one's most serious competitor is the bidder among the others with the highest signal.

In fact, all of the "order statistics" of the others' signals will come into play. By symmetry, let's always focus on bidder 1. From his perspectives, let $Y_1, Y_2, \ldots, Y_{N-1}$ denote the highest, second highest, ..., lowest of the other bidders' signals $S_2, \ldots, S_N$. From the seller's perspective, let $X_1, \ldots, X_N$ denote the highest, ..., lowest of all the signals $S_1, \ldots, S_N$.

5.1 Second Price Auction

5.1.1 Winner's Curse

Should a bidder with signal s_1 bid $\mathbb{E}[V_1|S_1 = s_1]$? This is *not* an equilibrium bidding function due to **winner's curse**: winning is bad news about your value since it means that all other bidders had signals below yours.

Formally, suppose all bidders bid according to $\tilde{b}(s) = \mathbb{E}[V_1|S_1 = s]$. By affiliation, $\tilde{b}(s)$ is strictly increasing in s, so the highest signal wins. So, if your signal is s, then

$$\begin{aligned}\mathbb{E}[V_1|S_1 = s] &= \mathbb{E}[V_1|S_1 = s, 0 \leq S_j < \infty, \forall j \neq 1] \\ &\geq \mathbb{E}[V_1|S_1 = s, 0 \leq S_j < s, \forall j \neq 1] = \mathbb{E}[V_1|S_1 = s \text{ wins}],\end{aligned}$$

where the inequality is often strict. Hence, there is (typically) a gap between your bid $\tilde{b}(s)$ and what the good is worth to you when you win. So if the second highest bid is between the gap, you will pay more than the good's worth (this is almost certainly the case when N is large). Bidding $\mathbb{E}[V_1|S_1 = s]$ is naive and non-strategic.

5.1.2 Equilibrium Behavior

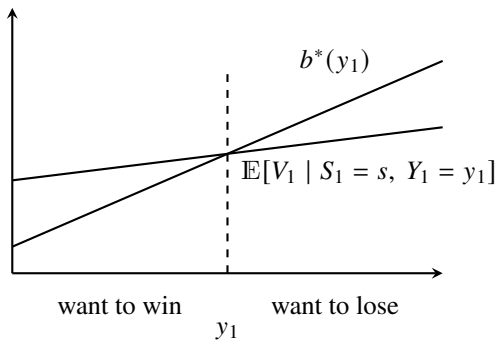
Note first that each bidder maximizes their expected payoff at the end of the auction, with respect to belief updated using any information revealed by the auction.

[4] Note that without symmetry, affiliation is not sufficient for these conclusions.

Let $b^*(s)$ denote the bidding function that all bidders use, and suppose it is strictly increasing. Denote the highest signal of others as y_1 and your signal as s. We can write the price and expected value as functions of y_1 in the following way:

$$\text{Price} = b^*(y_1),$$
$$\text{Expected Value} = \mathbb{E}[V_1|S_1 = s, Y_1 = y_1].$$

Both functions are nondecreasing in y_1 (price by assumption and expected value by affiliation). Now *suppose* the expected value function intersects once b^* from above:



You want to win if $\mathbb{E}[V_1|S_1 = s, Y_1 = y_1] > b^*(s)$ and lose otherwise. Bidding the (vertical axis) value of the intersection point achieves exactly this, no matter what y_1 is. Since the bidder bids using b^*, the intersection is exactly $(s, b^*(s))$. Thus we have:

Proposition 5.4. *In a SPA with interdependent values, an symmetric equilibrium bidding function b^* is*

$$b^*(s) = \mathbb{E}[V_1|S_1 = s, Y_1 = s].$$

The analysis above suggest only that b^* is a *candidate* equilibrium bidding function. To verify that it is indeed an equilibrium, we verify the assumptions made previously:

- $b^*(s) = \mathbb{E}[V_1|S_1 = s, Y_1 = s]$ is indeed strictly increasing in s by affiliation.
- $y_1 \mapsto \mathbb{E}[V_1|S_1 = s, Y_1 = y_1]$ indeed intersects $b^*(s)$ from above just once, due to affiliation.

5.1.3 Comparison with Rational Valuation

Note that $b^*(s) = \mathbb{E}[V_1|S_1 = s, Y_1 = s]$ is *not* a rational estimate for your value given signal s (why are you conditioning, after all, on an event that happens with measure 0). *Strategic bidding is not the same as rational valuation.*

In particular, suppose $s > y_1$ so $b^*(s)$ is a winning bid. Then,

$$\begin{aligned} b^*(s) = \mathbb{E}[V_1|S_1 = s, Y_1 = s] &\geq \underbrace{\mathbb{E}[V_1|S_1 = s, Y_1 = y_1]}_{\text{correct value estimate give } s,\, y_1} \\ &> \mathbb{E}[V_1|S_1 = y_1, Y_1 = y_1] = b^*(y_1). \end{aligned}$$

The right hand side is the price paid, which should be and is less than the value estimate given s, y_1 (and turns out to strictly so). There is no winner's curse.

Now suppose instead $s < y_1$, so $b^*(s)$ is a losing bid. Then,

$$\begin{aligned} b^*(s) = \mathbb{E}[V_1|S_1 = s, Y_1 = s] \leq \ & \underbrace{\mathbb{E}[V_1|S_1 = s, Y_1 = y_1]}_{\text{correct value estimate given } s,\, y_1} \\ & < \mathbb{E}[V_1|S_1 = y_1, Y_1 = y_1] = b^*(y_1). \end{aligned}$$

The right hand side is the winning bid. While it is true that you are bidding below your value estimate given s, y_1, note that you only win if your bid exceeds $b^*(y_1)$, which is higher than that value estimate. There is no loser's curse.

We thus have the following:

Proposition 5.5. *In a SPA with interdependent values:*

$$\textit{Losing bids} \ \leq \textit{correct value estimate} \leq \textit{winning bids}.$$

5.1.4 Seller Private Information

Suppose now that the seller has private information, summarizes say by a signal S_0.

Assumption 5.6. $S_0, S_1, \ldots, S_N, V_1, \ldots, V_N$ are affiliated and $S_0, S_1, \ldots, S_N$ are symmetric.

Question 5.7. Should the seller reveal or conceal her information?

Let's consider two possibilities: Either the seller commits to always revealing S_0 or to always concealing S_0.

(i) If the seller always conceals all information, then the equilibrium is as before, with $b^*(s) = \mathbb{E}[V_1|S_1 = s, Y_1 = s]$.

(ii) If the seller always reveals all information, we can repeat the same analysis as before to conclude

$$b^*(s, s_0) = \mathbb{E}[V_1|S_1 = s, Y_1 = s, S_0 = s_0].$$

(This is the same equilibrium bidding function as before, with the value-signal distribution conditional on $S_0 = s_0$.)

By the law of iterated expectations, we have

$$\begin{aligned} b^*(s) &= \mathbb{E}[V_1|S_1 = s, Y_1 = s] \\ &= \mathbb{E}_{S_0}\left[\mathbb{E}[V_1|S_1 = s, Y_1 = s, S_0]\middle|S_1 = s, Y_1 = s\right] \\ &= \mathbb{E}_{S_0}\left[b^*(s, S_0)\middle|S_1 = s, Y_1 = s\right]. \end{aligned}$$

Now, let R_U denote the expected revenue when the bidder are always uninformed of S_0, let R_I denote the expected revenue when the bidders are always informed of S_0, and let $X_1 \geq \cdots \geq X_N$ be the order statistics of $S_1, \ldots, S_N$.

We have

$$
\begin{aligned}
R_U &= \mathbb{E}\left[b^*(X_2)\right] \\
R_I &= \mathbb{E}\left[b^*(X_2, S_0)\right] = \mathbb{E}_{X_2}\left(\mathbb{E}_{S_0}[b^*(X_2, S_0)|X_2]\right).
\end{aligned}
$$

Thus,

$$
R_I - R_U = \mathbb{E}_{X_2}\left[\mathbb{E}_{S_0}[b^*(X_2, S_0)|X_2] - b^*(X_2)\right].
$$

We claim that $b^*(X_2) \leq \mathbb{E}_{S_0}[b^*(X_2, S_0)|X_2]$ for any X_2, say $X_2 = s$, and so the seller has incentive to reveal her information.

Note that

$$
\begin{aligned}
b^*(s) &= \mathbb{E}_{S_0}\left[b^*(s, S_0)|S_1 = s, Y_1 = s\right] \\
&= \mathbb{E}_{S_0}\left[b^*(s, S_0)|S_1 = X_2 = s, Y_1 = s\right] \\
&\leq \mathbb{E}_{S_0}\left[b^*(s, S_0)|S_1 = X_2 = s, Y_1 \geq s\right] \\
&= \mathbb{E}_{S_0}\left[b^*(s, S_0)|S_1 = X_2 = s\right] \\
&= \mathbb{E}_{S_0}\left[b^*(s, S_0)|S_1 = X_2 = s\right] = \mathbb{E}_{S_0}\left[b^*(s, S_0)|X_2 = s\right],
\end{aligned}
$$

where the fourth line follows from noting that $S_1 = S_2 = s$ implies that $Y_1 \geq s$, and the last line follows from symmetry.

Remark 5.8 (Intuition for why seller should always reveal/conceal). Consider the conceal case. Suppose seller's revenue comes from bidder 1, who bid $b^*(s) = \mathbb{E}[V_1|S_1 = s, Y_1 = s]$. This bid, being a losing bid, underestimates the true value V_1. A better estimate of V_1 is $E[V_1|S_1 = s, Y_1 \geq s] > b^*(s)$.

By revealing information S_0, the seller induces bidder 1 to bid $b^*(s, S_0) = \mathbb{E}[V_1|S_1 = s, Y_1 = s, S_0]$. This information helps bidder 1 correct (in expectation) her underestimation of V_1, thus increasing the seller's revenue. ☕

5.1.5 Optimal Disclosure

Consider first the class of disclosure policies where the seller reveals her signal S_0 if and only if it passes a threshold s^*. But in the case of no disclosure ($S_0 < s^*$), the bidders know that $S_0 < s^*$. This is equivalent to the case where S_0 follows the conditional distribution given $S_0 < s^*$, and we've already shown that full disclosure is optimal in this case. The same argument shows that is it never optimal conceal (fully or partially) any information.

That is,

Proposition 5.9. *In a SPA with interdependent values and seller private information, the seller should always reveal all information she has.*

5.2 English Auction

In the correlated private information model, the English auction is no longer equivalent to the second-price auction. Bidders gain information from the interdependence of values during the auction as the price rises and bidders drop out.

We will suppose by symmetry that all bidders use the same bidding strategy, described by $N-1$ functions

$$(b_0^*, \ldots, b_{N-2}^*),$$

where $b_0^*(s)$ is the drop-out price for a bidder with signal s when no one has yet dropped out, $b_1^*(s|p_1)$ is the drop-out price if exactly one other bidder has dropped out at price p_1, and

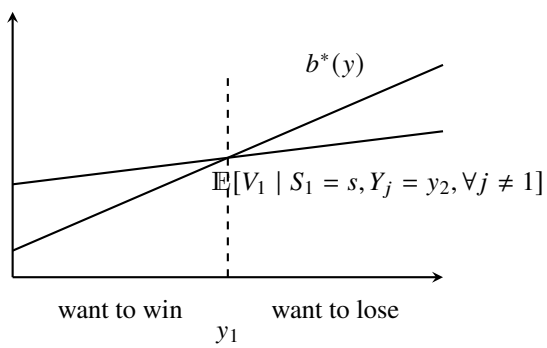
$$p_k^*(s|p_1, \ldots, p_k)$$

is the drop-out price if exactly k other bidders have dropped out at prices $p_1 < p_2 < \cdots < p_k$. Suppose each of these functions is strictly increasing in s.

Let's derive these functions, starting from b_0^*. The only event that matters for b_0^* is that of everyone else dropping out at the same price, causing you to win (otherwise you'd just jump to using the next function b_1^*). Thus suppose all other bidders have the same signal y and hence drop out at the same price $b_0^*(y)$. As before:

$$\begin{aligned} \text{Price} &= b_0^*(y), \\ \text{Value} &= \mathbb{E}[V_1|S_1 = s, Y_j = y, \forall j \neq 1], \end{aligned}$$

with price strictly increasing in y by assumption and expected value nondecreasing in y by affiliation. Suppose as before that the value function intersects $b_0^*(y)$ from above.



The same reasoning as before gives a candidate equilibrium bidding function:

$$b_0^*(s) = \mathbb{E}[V_1|S_1 = s, Y_j = s, \forall j].$$

From the affiliation assumption, we can verify that $b_0^*(s)$ is strictly increasing and that the value function crosses $b_0^*(s)$ from above just once.

Next, we consider $b_1^*(s|p_1)$. From the first drop out price p_1 we can infer the signal of the bidder who dropped out, $Y_{N-1} = y_{N-1} := (b_0^*)^{-1}(p_1)$, and use this to update our value estimate. As before, consider the case where all remaining bidders have the same signal $y = Y_1 = Y_2 = \cdots = Y_{N-2}$ and hence drop out at the same price $b_1^*(y|p_1)$:

$$\begin{aligned} \text{Price} &= b_1^*(y|p_1), \\ \text{Value} &= \mathbb{E}[V_1|S_1 = s, Y_1 = \cdots = Y_{N-2} = y, Y_{N-1} = y_{N-1}]. \end{aligned}$$

The same reasoning as before gives:

$$b_1^*(s|p_1) = \mathbb{E}[V_1|S_1 = s, Y_1 = \cdots = Y_{N-2} = s, Y_{N-1} = y_{N-1}],$$
$$y_{N-1} := (b_0^*)^{-1}(p_1).$$

This is the same bidding function as before, with the value-signal distribution updated using the information revealed by the first drop-out.

Repeating the same argument, we conclude the following:

Proposition 5.10. *In an English auction with interdependent values, an symmetric equilibrium bidding strategy* $(b_0^*, \ldots, b_{N-2}^*)$ *is given by*

$$b_k^*(s|p_1, \ldots, p_k) = \mathbb{E}[V_1|S_1 = Y_1 = \cdots = Y_{N-k-1} = s, Y_{N-k} = y_{N-k}, \ldots, Y_{N-1} = y_{N-1}],$$

where $y_{N-1}, \ldots, y_{N-k}$ *are such that*

$$\begin{aligned} b_0^*(y_{N-1}) &= p_1, \\ b_1^*(y_{N-2}|p_1) &= p_2, \\ &\vdots \\ b_{k-1}^*(y_{N-k}|p_1, \ldots, p_{k-1}) &= p_k. \end{aligned}$$

Thus, for instance,

$$\begin{aligned} b_0^*(s) &= \mathbb{E}[V_1|S_1 = s, Y_j = s, \forall j], \\ b_1^*(s|p_1) &= \mathbb{E}[V_1|S_1 = s, Y_1 = \cdots = Y_{N-2} = s, Y_{N-1} = y_{N-1}], \quad y_{N-1} := (b_0^*)^{-1}(p_1). \end{aligned}$$

Proof. Suppose your signal is s and let the others' signals be $y_1 \geq \cdots \geq y_{N-1}$, and suppose all others use the candidate equilibrium bidding strategy described above. To win, you'll have to stay in the auction until the price reaches at least

$$b_{N-2}^*(y_1|p_1, \ldots, p_{N-2}) = \mathbb{E}\big[V_1|S_1 = Y_1 = y_1, \underbrace{Y_2 = y_2, \ldots, Y_{N-1} = y_{N-1}}_{N-2 \text{ revealed signals}}\big],$$

which is exactly the drop-out price of the other bidder with the highest signal y_1.

The worth of the good, given the signals $y_1 \geq \cdots \geq y_{N-1}$ that will eventually be revealed, is

$$\mathbb{E}[V_1|S_1 = s, Y_1 = y_1, \ldots, Y_{N-1} = y_{N-1}].$$

You want to win if and only if this is greater than the price you pay if you win; that is, when $s > y_1$. Since the bidding functions are strictly increasing (and thus the winner is the bidder with the highest signal), you can achieve this by using them. □

Proposition 5.11. *When another bidder drops out, you revise your drop-out price downwards weakly, but never equal or below the current clock-price. Thus, the only way a bidder loses is when the clock-price exceeds their latest drop-out price, and never because more information revealed by other bidders.*

Proof. To see this, suppose you don't have the lowest signal, so $y_{N-1} < s$. Then,

$$\begin{aligned} p_1 = b_0^*(y_{N-1}) &= \mathbb{E}\left[V_1 | S_1 = Y_1 = \cdots = Y_{N-1} = y_{N-1}\right] \\ &< \mathbb{E}\left[V_1 | S_1 = Y_1 = \cdots = Y_{N-2} = s, Y_{N-1} = y_{N-1}\right] = b_1^*(s|p_1) \\ &\leq \mathbb{E}\left[V_1 | S_1 = Y_1 = \cdots = Y_{N-1} = s\right] = b_0^*(s), \end{aligned}$$

giving

$$b_0^*(s) \geq b_1^*(s|p_1) > p_1.$$

In general, if $y_{N-1} < y_{N-2} < \cdots < y_{N-k} < s$, then

$$\begin{aligned} p_k &= b_{k-1}^*(y_{N-k}|p_1, \ldots, p_{k-1}) \\ &= \mathbb{E}\big[V_1 | S_1 = Y_1 = \cdots = Y_{N-k} = y_{N-k}, \underbrace{Y_{N-k+1} = y_{N-k+1}, \ldots, Y_{N-1} = y_{N-1}}_{k-1 \text{ known signals}}\big] \\ &< \mathbb{E}\big[V_1 | S_1 = Y_1 = \cdots = Y_{N-k-1} = s, \underbrace{Y_{N-k} = y_{N-k}, \ldots, Y_{N-1} = y_{N-1}}_{k \text{ known signals}}\big] = b_k^*(s|p_1, \ldots, p_k) \\ &\leq \mathbb{E}\big[V_1 | S_1 = Y_1 = \cdots = Y_{N-k} = s, \underbrace{Y_{N-k+1} = y_{N-k+1}, \ldots, Y_{N-1} = y_{N-1}}_{k-1 \text{ known signals}}\big] = b_{k-1}^*(s|p_1, \ldots, p_{k-1}), \end{aligned}$$

giving

$$b_{k-1}^*(s|p_1, \ldots, p_{k-1}) \geq b_k^*(s|p_1, \ldots, p_k) > p_k.$$

□

5.3 Revenue Comparison between SPA and EA

Let $X_1 \geq \cdots \geq X_n$ be the order statistics of $S_1, \ldots, S_N$.

Consider a fictitious auction (impossible to implement), called SPA_2, defined as follows: The seller observes all but the two highest signals and reveals the signals she observes. So, she observes and reveal $X_3 = x_3, \ldots, X_N = x_N$ and then allows the two bidders with the highest signals to bid in a SPA.

The equilibrium strategy is

$$b_{\text{SPA}_2}^*(s|x_3, \ldots, x_N) = \mathbb{E}[V_1 | S_1 = Y_1 = s, Y_2 = x_3, \ldots, Y_{N-1} = x_N].$$

So, if $X_1 = x_2 \geq X_2 = x_2 \geq \cdots \geq X_n = x_n$, then

$$R_{\text{SPA}_2}(x_1, \ldots, x_n) = \mathbb{E}[V_1 | S_1 = Y_1 = x_2, Y_2 = x_3, \ldots, Y_{N-1} = x_n] = R_{\text{EA}}(x_1, \ldots, x_n).$$

Thus SPA_2 has the same expected revenue as the EA. But by our previous analysis on information revelation,

$$R_{\text{EA}} = R_{\text{SPA}_2} \geq R_{\text{SPA}}.$$

Question 5.12. Why are SPA used then? A potential answer: we did not model risk aversion.

Proposition 5.13. *With interdependent values, $R_{EA} \geq R_{SPA}$.*

5.4 Efficiency of SPA and EA

Note that while the highest signal wins, it is not necessarily the case that the highest value wins.

Proposition 5.14. *With interdependent values, the SPA and EA are not efficient.*

For an explicit example, consider the following case:

$$V_1 = S_1 + 2S_2, \quad V_2 = 2S_1 + S_2.$$

In a SPA, it is essay to verify that $b^*(s) = 3s$. Then, if $S_1 = 5$ and $S_2 = 2$, we have bidder 1 winning, but $V_1 < V_2$.

5.5 First-Price Auction

Let $f_{Y_1}(y|s)$ be the probability density that the highest signal of the others is y given that your signal is s. By affiliation, $f_{Y_1}(\overline{y}|s)/f_{Y_1}(\underline{y}|s)$ is nondecreasing in s for each $\overline{y} \geq \underline{y}$. This implies that

Proposition 5.15. *$f_{Y_1}(y|s)/F_{Y_1}(y|s)$ is nondecreasing in s for each y.*

To show this, write F as an integral of f and check the reciprocal.

Now define

$$w(s, y) \coloneqq \mathbb{E}[V_1 | S_1 = s, Y_1 = y]$$

as the expected value of the good to you if you win and the highest other signal is y. Let $\hat{b}(\cdot)$ denote a strictly increasing symmetric equilibrium bidding function.

Let $u(r, s)$ denote your expected utility when your signal is s and you bid $\hat{b}(r)$ and all bidders bid according to $\hat{b}$. Then, so long as $\hat{b}(\cdot)$ is strictly increasing (which we are assuming but must later verify), we have

$$u(r, s) = \mathbb{E}\left[[w(s, y) - \hat{b}(r)]\mathbb{1}_{r>y} \mid s\right] = \int_0^r [w(s, y) - \hat{b}(r)] f_{Y_1}(y|s)\, \mathrm{d}y.$$

The optimality condition implies that $\partial_r u(r, s)|_{r=s} = 0$ for each s, or

$$\begin{aligned} 0 &= \left[-\int_0^r \hat{b}'(r) f_{Y_1}(y|s)\, \mathrm{d}y + \left[w(s, r) - \hat{b}(r)\right] f_{Y_1}(r|s) \right]_{r=s} \\ &= \left[w(s, s) - \hat{b}(s)\right] f_{Y_1}(s|s) - \int_0^s \hat{b}'(s) f_{Y_1}(y|s)\, \mathrm{d}y \\ &= \left[w(s, s) - \hat{b}(s)\right] f_{Y_1}(s|s) - \hat{b}'(s) F_{Y_1}(s|s). \end{aligned}$$

Thus,

$$\left[w(s, s) - \hat{b}(s)\right] f_{Y_1}(s|s) = \hat{b}'(s) F_{Y_1}(s|s).$$

Note that the left side is the marginal benefit of increasing the bid, while the right side is the marginal cost of increasing the bid. We can rewrite this to

$$\hat{b}'(s) = \left[w(s, s) - \hat{b}(s)\right] \frac{f_{Y_1}(s|s)}{F_{Y_1}(s|s)}, \quad \forall s.$$

Solving this ODE with the boundary condition $\hat{b}(0) = w(0,0)$ (argue from first principles) gives the following candidate equilibrium bidding function:

Proposition 5.16. *In a FPA with interdependent values, a symmetric equilibrium bidding function is given by*

$$\hat{b}(s) = \int_0^s w(y,y)h(y|s)\,\mathrm{d}y, \quad \forall s > 0$$
$$\hat{b}(0) = w(0,0),$$

where

$$h(y|s) := \frac{f_{Y_1}(y|y)}{F_{Y_1}(y|y)} \exp\left(-\int_y^s \frac{f_{Y_1}(x|x)}{F_{Y_1}(x|x)}\,\mathrm{d}x\right), \quad \forall y \le s.$$

That is this in fact an equilibrium will be verified later.

We check now that $\hat{b}$ is strictly increasing. Let H be the corresponding cdf of h:

$$H(y|s) := \int_0^y h(z|s)\,\mathrm{d}z = \exp\left(-\int_y^s \frac{f_{Y_1}(x|x)}{F_{Y_1}(x|x)}\,\mathrm{d}x\right), \quad \forall y \le s.$$

Note that $H(s|s) = 1$ and $H(0|s) = 0$. Note also that since $h(y|s)$ is positive on $y \in [0,s]$, $h(y|s)$ and $H(y|s)$ do indeed behave like conditional pdfs and cdfs on $[0,s]$.

Thus, $\hat{b}(s)$ is a weighted average of $w(y,y)$ for $y \in [0,s]$. Using the fact that $H(s|s) = 1$, we have

$$\hat{b}(s) = \int_0^s w(y,y)h(y|s)\,\mathrm{d}y < \int_0^s w(s,s)h(y|s)\,\mathrm{d}y = w(s,s), \quad \forall s > 0,$$

Thus,

Proposition 5.17. $\hat{b}(s) < w(s,s)$ *for all* $s > 0$.

Using this fact, and recalling

$$\hat{b}'(s) = \left[w(s,s) - \hat{b}(s)\right] \frac{f_{Y_1}(s|s)}{F_{Y_1}(s|s)}, \quad \forall s,$$

we have that $\hat{b}$ is indeed strictly increasing and so our previous formula for $u(r,s)$ is valid.

We now verify that $\hat{b}(\cdot)$ above is a symmetric equilibrium (we've assumed $\hat{b}(\cdot)$ is a equilibrium and derived it using the implied FOC; we now need to check that SOC holds). Recall that

$$\begin{aligned}\frac{\mathrm{d}u(r,s)}{\mathrm{d}r} &= \left[w(s,r) - \hat{b}(r)\right] f_{Y_1}(r|s) - \hat{b}'(r)F_{Y_1}(r|s) \\ &= F_{Y_1}(r|s)\left[(w(s,r) - \hat{b}(r))\frac{f_{Y_1}(r|s)}{F_{Y_1}(r|s)} - \hat{b}'(r)\right].\end{aligned}$$

We want to show that the derivative is positive for $r < s$ and negative for $r > s$. Instead of starting from $r = s$ and move r, we will move s. When we decrease s so that $s < r$, note that $w(s,r)$ decreases and $f_{Y_1}(r|s)/F_{Y_1}(r|s)$ weakly decreases, so the derivative becomes negative. The other direction of perturbation is similar.

5.6 Revenue Comparison of FPA, SPA, and EA

It will be useful to note the following:

Lemma 5.18. *$f_{Y_1}(y|s)/F_{Y_1}(s|s)$ is the probability density that the second highest signal X_2 equals y given the highest signal X_1 is s.*

Proof. $f_{Y_1}(y|s)$ is the probability distribution of Y_1 given $S_1 = s$. Thus, $f_{Y_1}(y|s)/F_{Y_1}(a|s)$ is the probability distribution of $Y_1 = y$ given $S_1 = s$ and $Y_1 \leq a$.

In particular, $f_{Y_1}(y|s)/F_{Y_1}(s|s)$ is the probability distribution of $Y_1 = y$ given $S_1 = s$ and $Y_1 \leq s$, which is the density of $X_2 = y$ given $S_1 = X_1 = s$. We conclude by symmetry that this is also the density of $X_2 = y$ given $X_1 = s$. □

Recall that we showed $R_{\text{SPA}} \leq R_{\text{EA}}$. We now argue that $R_{\text{FPA}} \leq R_{\text{SPA}}$. In a SPA,

$$b^*(s) = \mathbb{E}[V_1|S_1 = s, Y_1 = s] = w(s, s).$$

Let $R|_{X_1=s}$ denote the expected revenue given that the highest signal X_1 is s. Clearly,

$$R_{\text{FPA}}|_{X_1=s} = \hat{b}(s) = \int_0^s w(y, y)h(y|s)\,\mathrm{d}y, \quad H(y|s) = \exp\left(-\int_y^s \frac{f_{Y_1}(x|x)}{F_{Y_1}(x|x)}\,\mathrm{d}x\right),$$
$$R_{\text{SPA}}|_{X_1=s} = \int_0^s w(y, y)\frac{f_{Y_1}(y|s)}{F_{Y_1}(s|s)}\,\mathrm{d}y$$

We will show that $R_{\text{FPA}}|_{X_1=s} \leq R_{\text{SPA}}|_{X_1=s}$ for each s, which implies $R_{\text{FPA}} \leq R_{\text{SPA}}$. Since $w(y, y)$ is strictly increasing in y, and since $\hat{b}(s)$ is a weighted average of $w(y, y)$, $y \in [0, s]$, it suffices to show that $h(y|s)$ puts more weight on lower values of y than does $f_{Y_1}(y|s)/F_{Y_1}(s|s)$, or, equivalently, that $H(y|s) \geq F_{Y_1}(y|s)/F_{Y_1}(s|s)$. So we will be content to show that

$$H(y|s) \geq \frac{F_{Y_1}(y|s)}{F_{Y_1}(s|s)}, \quad \forall y \leq s.$$

To see this inequality, note that

$$\begin{aligned} H(y|s) &= \exp\left(-\int_y^s \frac{f_{Y_1}(x|x)}{F_{Y_1}(x|x)}\,\mathrm{d}x\right) \geq \exp\left(-\int_y^s \frac{f_{Y_1}(x|s)}{F_{Y_1}(x|s)}\,\mathrm{d}x\right) \\ &= \exp\left(-\log F_{Y_1}(x|s)\Big|_y^s\right) = \exp\left(-\log\left(\frac{F_{Y_1}(s|s)}{F_{Y_1}(y|s)}\right)\right) = \frac{F_{Y_1}(y|s)}{F_{Y_1}(s|s)}. \end{aligned}$$

Thus we have

Proposition 5.19. *In an interdependent values mode, $R_{FPA} \leq R_{SPA} \leq R_{EA}$.*

Remark 5.20. The first inequality is driven by correlated values, the second by information revealed during the auction. ☕

6 Markets as Auctions I: Independent Private Values

- m homogeneous units of an indivisible good for sale.
- N bidders, each with single unit demand. Bidders have independent private values $v_i \overset{\text{iid}}{\sim} f_i$ for the good.
- Bidders submits (potentially untruthful) demand functions/bids $b_1, \ldots, b_N$, whose order statistics are denoted $B_{(1)} \geq \cdots \geq B_{(N)}$.

Use $B^{-i}_{(k)}$ to denote the kth highest bid among the bids of all bidders except i. The range of market clearing prices given the bids is $[B_{(m+1)}, B_{(m)}]$. Suppose the lowest market-clearing price, $B_{(m+1)}$, is always chosen as the market clearing price.

Consider an **$(m+1)$st-price auction (m units)** (also a special case of a **uniform-price auction**): each bidder submit a (sealed) bid, the m highest bids win a unit, and each winner pays the $(m+1)$st highest bid.

Proposition 6.1. *It is a dominant strategy to bid your value in a $(m+1)$st-price auction when bidders have independent private values. Thus the outcome is efficient.*

Proof. If bidder i wins, she pays $B^{-i}_{(m)}$. She thus wants to win when $v_i > B^{-i}_{(m)}$ and lose otherwise. This can be achieved by bidding exactly $b_i = v_i$. □

7 Markets as Auctions II: Interdependent Values

Let's return to the general symmetric interdependent values affiliated private information model:

- $m \geq 1$ homogeneous units of an indivisible good for sale.
- N bidders with single unit demand. Values: $V_1, \ldots, V_N$. Signals: $S_1, \ldots, S_N$.

How should bidders bid in an $(m+1)$st-price auction. Let $b^*_{N,m}(s)$ be a symmetric equilibrium bidding function. Suppose $y_1 > y_2 > \cdots > y_{N-1}$ are the order statistics of the other bidders' signals. The price you pay if you win is $b^*_{N-m}(y_m)$. The good's worth to you if you win given your signal is s is

$$\mathbb{E}\left[V_1 | S_1 = s, Y_m = y_m\right].$$

The same analysis as before shows that

Proposition 7.1. *In an $(m+1)$st-price auction with interdependent values, a symmetric equilibrium bidding strategy $b^*_{N,m}(\cdot)$ is given by*

$$b^*_{N,m}(s) = \mathbb{E}\left[V_1 | S_1 = s, Y_m = s\right].$$

7.1 Common Values

We now specialize the model to the case of **common values** and ask: if the market is sufficiently large (N, m are large), will the market price reflect the true value of the good?

Setup:

- Common values $V_1 = \ldots V_N = V \sim g(v)$, with values in $[0,1]$, (think stocks, IPOs, Treasury bills).
- N bidders with single unit demand.
- For each fixed value $V = v \in [0,1]$, each bidder receives an independent signal $S_i \sim \text{Uniform}(0, v)$.

Now suppose $N, m \to \infty$ such that $m/N \to \alpha \in (0,1)$. Since $s \sim \text{Uniform}(0, v)$, approximately αN of the N bidders will have signals above $(1-\alpha)v$, so the $(m+1)$th highest signal X_{m+1} will be approximately $(1-\alpha)v$. Now for any bidder whose signal is s,

$$b^*_{N,m}(s) = \mathbb{E}\left(V_1 | S_1 = s, Y_m = s\right) \approx \mathbb{E}\left(V_1 | X_{m+1} = s\right) \approx \frac{s}{1-\alpha}.$$

The equilibrium price is

$$P_{N,m} = b^*_{N,m}(X_{m+1}) \approx \frac{X_{m+1}}{1-\alpha} \longrightarrow \frac{(1-\alpha)v}{1-\alpha} = v.$$

8 Multi-Unit Demand

Setup:

- m units for sale.
- N bidders.
- Private values for bidder i of the m units are $v_i = (v_{i1}, \ldots, v_{im})$, which we assume to be weakly decreasing in the unit index (marginal values decrease). Thus v_{ik} is i's marginal value for the kth unit.
- If bidder i wins k units and pays a total price of P, her utility is $v_{i1} + \cdots + v_{ik} - P$.
- v_i is drawn from $[0, 1]^m$ according to $f_i(v_i)$. The value profiles $v_1, \ldots, v_N$ are independent.

We will be concerned with the following questions:

(i) What are the analogues of FPA, SPA, EA in this case?[5]

(ii) Are there auctions that allocate the m units efficiently (ex post, would any voluntary exchange happen)?

8.1 Generalizations of SPA

Consider the $(m + 1)$st-price auction as a generalization of the single-unit SPA. Each bidder i submits $b_i = (b_{i1}, \ldots, b_{im})$ with $b_{11} \geq \cdots \geq b_{1m}$. If bidder i wins k units, she pays the $(m + 1)$st-highest bid for each unit won. So she pays $B_{(m+1)} \cdot k$.

This is not efficient:

Example 8.1. $N = m = 2$, $v_1 = (10, 10)$, $v_2 = (7, 0)$.

If each bids truthfully, bidder 1 wins both units and pays 7 for each unit, gaining utility 6. She thus would deviate to say $(10, 0)$, getting utility 10.

In fact, $(10, 0)$ and $(7, 0)$ is an equilibrium, and it is *inefficient*: bidder 1 and 2 each wins one unit, but bidder 1 values the second unit more than bidder 2 does so the first good.

Let's give the SPA another shot. Perhaps the problem is that a player's bid can influence her payment. Let's consider then a sequential SPA, where the m units are auctioned off sequentially in m rounds, and in each round the highest bidder wins and pays the second highest bid.

Example 8.2. $N = m = 2$, $v_1 = (10, 10)$, $v_2 = (7, 0)$.

Fist consider the case where bidders bid truthfully. In the first SPA, the bidders bid 10 and 7 respectively, so 1 wins and pays 7. In the second SPA, the bidders bid 10 and 7, so 1 wins and pays 7 again. This is an efficient outcome but not an equilibrium: bidder 1 can deviate to bidding 0 in the first SPA and 10 in the second, winning the second unit and paying 0 for it, thus gaining utility 10 instead of 6.

[5] Dutch is less common in the multi-unit case.

8.2 Ascending Bid Auction (Price Clocks)

Consider then a generalization of the EA with multiple price clocks.

Consider the two goods case and call them good A and good B. We have separate price clocks for A and B, with prices p_A and p_B starts at 0 and each price increases at rate 1 if the demand for that good exceeds 1.

Example 8.3. $N = m = 2$, $v_1 = (10, 10)$, $v_2 = (7, 0)$. When bidders bid truthfully, bidder 2 "chatters" between A and B, whichever is cheaper. Both of the prices goes up at the same rate, until both reach 7, at which point bidder 2 drops out and the clock stops (since there is no excess demand) and bidder 1 wins both goods, paying 7 for each.

This is again not an equilibrium: bidder 1 can instead demand only one good say A up to 10. Then the auction ends immediately with each bidder winning one unit and paying 0. The equilibrium outcome is again inefficient.

A similar analysis shows that even with the same price clock for both goods, the outcome is still inefficient.

8.3 Discriminatory Auction (Generalized Sealed-Bid FPA)

A **discriminatory auction** proceeds as follows: Each bidder submits a vector of bids $b_i = (b_{i1}, \ldots, b_{im})$ with $b_{i1} \geq \cdots \geq b_{im}$. The highest m bids win, with winners paying their bids on each unit won. So if i wins k units, she has utility

$$u_i = v_{i1} + \cdots + v_{ik} - (b_{i1} + \cdots + b_{ik}).$$

Let $(\hat{b}_1(\cdot), \ldots, \hat{b}_m(\cdot))$ denote a symmetric equilibrium bidding strategy for DA. So

$$\hat{b}_k(v_{i1}, \ldots, v_{im})$$

is the bid on a kth unit submitted by a bidder with values $v_{i1}, \ldots, v_{im}$. Suppose $\hat{b}_k$ is strictly increasing in v_{ij} for each j. One can show that such an equilibrium exists (we will not be bothered with the details).

Is this outcome efficient?

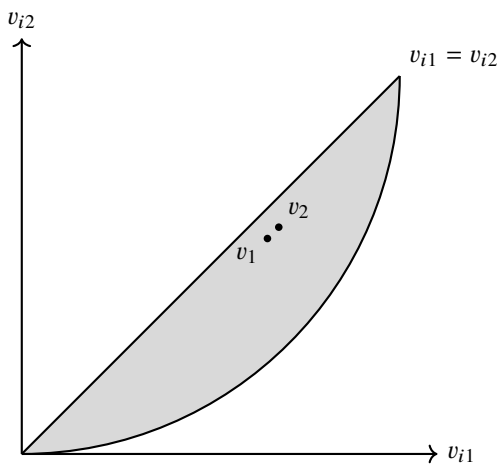
Example 8.4. Consider again $m = N = 2$. Note that

- 1 wins at least one unit if and only if $b_{11} > b_{22}$, and similarly for 2.
- 1 wins both units if and only if $b_{12} > b_{21}$ and similarly for 2.

(Visualize this as a cross.)

If $v_{i1} \approx v_{i2}$, because i's bid for a 1st unit is against a lower bid distribution (that of $\hat{b}_2(V_{j1}, V_{j2})$) than is i's bid for a second unit (against the distribution of $\hat{b}_1(V_{j1}, V_{j2})$). Thus ideally i would want $b_{i1} < b_{i2}$ but this is not possible. Therefore the optimal bid vector for i is to set $\hat{b}_1(v_{i1}, v_{i2}) = \hat{b}_2(v_{i1}, v_{i2})$.

Now consider a figure of v_{i1} against v_{i2}.



Note that only the area below the diagonal $v_{i1} = v_{i2}$ is relevant. In some region near the diagonal $v_{i1} = v_{i2}$, we have $\hat{b}_1(v_{i1}, v_{i2}) = \hat{b}_2(v_{i1}, v_{i2})$. Let $v_1 = (v_{11}, v_{12})$ and $v_2 = (v_{21}, v_{22})$ be two points in this region, with v_2 to the northeast of v_1. Then,

$$\hat{b}_1(v_1) = \hat{b}_2(v_1) < \hat{b}_1(v_2) = \hat{b}_2(v_2).$$

It is possible that

$$v_{21} > v_{11} > v_{22} > v_{12},$$

in which case the efficient outcome is for each bidder to win one unit, but in equilibrium, one bidder win both units.

The crucial fact here is that the high bids of each bidder is not pitted against the other high bids.

8.4 Vickrey's Auction

We want an auction that allocates the m units to the bidder(s) with the highest m values. If we can get the bidders to bid honestly, i.e., $b_i = (v_{i1}, \ldots, v_{im})$, then allocating the units to the m highest bids will be efficient.

So let's focus attention on sealed-bid auctions that allocate the units to the m highest bids. The only remaining feature of the auctions to determine, then, is what the winners' pay. Let's assume they pay only if they win some units.

Suppose $m = N = 2$. Write $v_1 = (v_{11}, v_{12})$ and suppose bidder 2 bids $b_2 = (b_{21}, b_{22})$. Can we define payment rule for bidder 1 so that 1 wins when he wants to win and loses when he wants to lose if he bids honestly? Denote the price bidder 1 pays if he wins a kth unit by p_{1k}.

Note that from bidder 1's perspective, we have the following:

	Wins	Wants to win
First unit	$v_{11} > b_{22}$	$v_{11} > p_{11}$
Second unit	$v_{12} > b_{21}$	$v_{12} > p_{12}$

This suggests setting

$$p_{11} \coloneqq b_{22}, \quad p_{12} \coloneqq b_{21}.$$

Bidder's 2 price can be found with a symmetric analysis. Thus, with two bidders, the pricing rule is

$$p_{ik} = \begin{cases} b_{j2}, & k = 1 \\ b_{j1}, & k = 2 \end{cases}.$$

We first generalize the number of units to m. Again suppose 1 bids $b_1 = v_1 = (v_{11}, \ldots, v_{1m})$ and similarly for 2. Note that bids are competing against each other in a cross pattern: 1's bid for a kth unit is competing against 2's bid for a $(m + 1 - k)$th unit. (E.g., to win the first unit, 1 has to bid higher than 2's bid for the mth unit.)

	Wins	Wants to win
First unit	$v_{11} > b_{2m}$	$v_{11} > p_{11}$
Second unit	$v_{12} > b_{2(m-1)}$	$v_{12} > p_{12}$
kth unit	$v_{1k} > b_{2(m+1-k)}$	$v_{1k} > p_{1k}$
mth unit	$v_{1m} > b_{21}$	$v_{1m} > p_{1m}$

Thus, we should set

$$p_{ik} = b_{2m-k+1}.$$

Let's check that this works. Suppose that $p_{ik} = b_{j,m-k+1}$ for $i, j = 1, 2, i \neq j$. Is it a dominant strategy to bid $b_i = (v_{i1}, \ldots, v_{im})$?

Suppose $b_i = (v_{i1}, \ldots, v_{im})$ and suppose i wins k units when j bids $b_j = (b_{j1}, \ldots, b_{jm})$. Then, the highest m bids are winning: $v_{i1}, \ldots, v_{ik}$ and $b_{j1}, \ldots, b_{j,m-k}$. To win a $(k + 1)$st unit, you'd need to pay $b_{j,m-k}$, but your value for that unit is $v_{i,k+1} < b_{j,m-k}$, so you don't want to win that unit. Do you want to lose a kth unit? Well, the kth unit is worth v_{ik} to you, and costs $b_{j,m-k+1} < v_{ik}$, so you want to win that unit.

Note that fixing $N = 2$ and $m = 1$, this is exactly the SPA.

8.5 Generalization to N Bidders

Again suppose i bids $b_i = v_i$ and suppose other bidders $j \neq i$ bid b_j. Let $B^{-i}_{(k)}$ denote the kth highest bid of the others.

Note that i wins the kth unit if and only if $v_{ik} > B^{-i}_{(m+1-k)}$ and they wants to win if and only if $v_{ik} > p_{ik}$. Thus, we should set

$$p_{ik} = B^{-i}_{(m+1-k)}.$$

Remark 8.5. Here, the payment is exactly the opportunity you caused to the other bidders by winning the kth unit. Your pay your externality on the other bidders. ☕

Definition 8.6. The following is called a **multi-unit (m unit) Vickrey auction**: All bidders submit m sealed bids. The m highest bids are winning. If a bidder wins k units, she pays the sum of the k highest losing bids of her opponents.

Theorem 8.7. *It is a dominant strategy to bid honestly in a multi-unit Vickrey. Hence, the outcome of rational bidding is efficient.*

9 Efficiency and Interdependent Values with Many units and Multiunit Demand

For this analysis we will specialize to $N = 2$ and m units (the analysis can be extended to more bidders). Each bidder $i = 1, 2$ receives a signal s_i. Let $v_{ik}(s_1, s_2)$ denote i's value for a kth unit when the signals $s_1, s_2 \in [0, \infty)$. For example, we may have

$$v_{ik}(s_1, s_2) = \mathbb{E}\left[V_{ik} | S_1 = s_1, S_2 = s_2\right].$$

Instead of building from affiliation, we now directly make assumption on $v_{ik}(s_1, s_2)$.

Assumption 9.1.

(A1) Nonincreasing marginal values: $v_{ik}(s_1, s_2) \geq v_{i,k+1}(s_1, s_2) \geq 0$.

(A2) $v_{ik}(s_1, s_2)$ is strictly increasing in s_i and nondecreasing in s_j.

(A3) Your signal matters more for your value than does the other person's signal:

$$\frac{\partial v_{1k}(s_1, s_2)}{\partial s_1} \geq \frac{\partial v_{2\ell}(s_1, s_2)}{\partial s_1}, \quad \frac{\partial v_{1k}(s_1, s_2)}{\partial s_2} \leq \frac{\partial v_{2\ell}(s_1, s_2)}{\partial s_2}.$$

Equivalently, $v_{1k}(s_1, s_2) - v_{2\ell}(s_1, s_2)$ is nondecreasing in s_1 and nonincreasing in s_2.

(A4) There exists unique α, β such that

$$v_{1k}(s_1, \alpha) = v_{2\ell}(s_1, \alpha), \quad v_{1k}(\beta, s_2) = v_{2\ell}(\beta, s_2), \quad \forall k, \ell.$$

Our goal: show that Vickrey's sealed-bid auction has an efficient equilibrium. Suppose

$$\hat{b}_1(s_1) = \left(\hat{b}_{11}(s_1), \ldots, \hat{b}_{1m}(s_1)\right)$$

and similarly for $\hat{b}_2(s_2)$ are equilibrium bidding strategies. Assume all $\hat{b}_{ik}(s_i)$ are strictly increasing in s_i.

We will use the assumptions above to determine what they are. Recall that $\hat{b}_{1k}(s_1)$ competes against $\hat{b}_{2,m+1-k}(s_2)$, which is also the marginal price that 1 pays for a kth unit. Similarly, $\hat{b}_{1k}(s_1)$ is the marginal price that 2 pays for an $(m + k - 1)$st unit. Hence, 1's optimal bid for a kth unit depends on $\hat{b}_{2,m-k+1}(s_2)$ and 2's on $\hat{b}_{1k}(s_1)$. We must derive the two functions simultaneously.

Let s_1^0 be any signal of bidder 1. We will determine $\hat{b}_{1k}(s_1^0)$. Note that $\hat{b}_{2,m-k+1}(s_2)$ is the price 1 pays for her kth unit and $V_{1k}(s_1^0, s_2)$ is the value of that unit to her. Suppose as before that $s_2 \mapsto V_{1k}(s_1^0, s_2)$ crosses $s_2 \mapsto \hat{b}_{2,m-k+1}(s_2)$ from above. You'd want to win when $V_{1k} > \hat{b}_{2,m-k+1}$ and lose otherwise, so the optimal bid is the height of the crossing point. But you don't know this point since you don't know s_2. This is related to the fact that the two bidding functions have to be determined jointly.

Say the crossing point occurs at s_2^0. Then, we have

$$\hat{b}_{1k}(s_1^0) = V_{1k}(s_1^0, s_2^0) = \hat{b}_{2,m-k+1}(s_2^0),$$

the first being bidder 1's bid, and the latter two being the height of the crossing point.

Symmetrically,

$$\hat{b}_{2,m-k+1}(s_2^0) = V_{2,m-k+1}(s_1^0, s_2^0) = \hat{b}_{1k}(s_1^0).$$

The interpretation of s_1^0 and s_2^0 is the following: s_2^0 is the breaking point blah blah blah.

These two displays can be summarized as

$$\hat{b}_{1k}(s_1^0) = V_{1k}(s_1^0, s_2^0) = V_{2,m-k+1}(s_1^0, s_2^0) = \hat{b}_{2,m-k+1}(s_2^0).$$

With this, assumption (A4) allows us to define the bidding functions $\hat{b}_{1k}(\cdot)$ and $\hat{b}_{2,m-k+1}(\cdot)$ for all s_1 and s_2.

How? Consider unit k and any signal s_1. The assumption gives a unique signal α for bidder 2 such that $V_{1k}(s_1, \alpha) = V_{2,m-k+1}(s_1, \alpha)$. We then (by the display above) define

$$\hat{b}_{1k}(s_1) = v_{1k}(s_1, \alpha).$$

Similarly, for any k and any signal s_2, we find the unique β such that $V_{1k}(\beta, s_2) = V_{2,m-k+1}(\beta, s_2)$, and define

$$\hat{b}_{2,m-k+1}(s_2) = v_{2,m-k+1}(\beta, s_2).$$

We now have candidate equilibrium bidding functions. We check that these functions yield an equilibrium and an efficient outcome.

Note that $\alpha(s_1, k)$ is weakly decreasing in k (since the RHS in $V_{1k}(s_1^0, \alpha) = V_{2,m-k+1}(s_1^0, \alpha)$ increases, and we need to decrease α to restore equality) and weakly increasing in s_1 (LHS increases more due to (A3) blah blah). Similarly, $\beta(s_2, k)$ is weakly decreasing in k and weakly increasing in s_2.

Thus, $\hat{b}_{ik}(s_1) = V_{ik}(s_1, \alpha)$ is strictly increasing in s_1 and nondecreasing in k.

Remains to show efficiency and verify equilibrium.

We start with efficiency. Let s_1^0, s_2^0 be private signals. Suppose that with these signals, it is efficient for bidder 1 to receive k units and for bidder 2 to receive $m - k$. Write $s^0 = (s_1^0, s_2^0)$. Then, the highest m values are $v_{11}(s^0), \ldots, v_{1k}(s^0)$ and $v_{21}(s^0), \ldots, v_{2,m-k}(s^0)$.

We have

(i) $v_{1k}(s_1^0, s_2^0) > v_{2,m-k+1}(s_1^0, s_2^0)$

(ii) $v_{2,m-k}(s_1^0, s^0) > v_{1k}(s_1^0, s_2^0)$.

We must show that if the two bidders use their $\hat{b}$ functions, then bidder 1 wins exactly k units. By definition,

$$\hat{b}_{1k}(s_1^0) = v_{1k}(s_1^0, \alpha) = v_{2,m-k+1}(s_1^0, \alpha),$$

so from (i), we have $\alpha(s_1^0, k) > s_2^0$. Thus,

$$\hat{b}_{1k}(s_1^0) = v_{1k}(s_1^0, \alpha) \geq v_{2,m-k+1}(s_1^0, s_2^0).$$

Symmetrically (still using (i)) $\beta < s_1^0$ and then

$$\hat{b}_{2,m-k}(s_2^0) = v_{2,m-k}(\beta, s_2^0) = v_{1k}(\beta, s_2^0) \leq v_{1k}(s_1^0, s_2^0).$$

Now, we have

$$\hat{b}_{1k}(s_1^0) \geq v_{1k}(s_1^0, s_2^0) > v_{2,m-k+1}(s_1^0, s_2^0) \geq \hat{b}_{2,m-k+1}(s_2^0).$$

On the other hand, using (ii), we conclude

$$\hat{b}_{2,m-k}(s_2^0) \geq v_{2,m-k}(s_1^0, s_2^0) > v_{1,k+1}(s_1^0, s_2^0) \geq \hat{b}_{1,k+1}(s_1^0).$$

Hence the outcome is efficient regardless of the signals.

Now we check that the bidding functions are indeed an equilibrium. Note that each of

$$v_{11}(s_1^0, s_2^0), \ldots, v_{1k}(s_1^0, s_2^0); \hat{b}_{21}(s_2^0), \ldots, \hat{b}_{2,m-k}(s_2^0)$$

are weakly larger than each of the following

$$v_{1,k+1}(s_1^0, s_2^0), \ldots, v_{1m}(s_1^0, s_2^0); \hat{b}_{2,m-k+1}(s_2^0), \ldots, \hat{b}_{2,m}(s_2^0).$$

Now, bidder 1 gets k units. Would she want to deviate from this? To win an extra $(k+1)$th unit, she has to pay $\hat{b}_{2,m-k}(s_2^0) > v_{1,k+1}(s_1^0)$, so she doesn't want to win that unit. To lose a kth unit, she has to give up a unit worth $v_{1k}(s_1^0, s_2^0)$ and save $\hat{b}_{2,m-k+1}(s_2^0) < v_{1k}(s_1^0, s_2^0)$.

For the same reason, bidder 1 would not want to deviate to win more/fewer than 2 units. Same for bidder 2.

So even in the interdependent values case, we can have an efficient equilibrium with Vickrey's sealed-bid auction.

Example 9.2. $m = 2$;

$$v_1 = 30s_1 + 20s_2, \quad v_2 = 20s_1 + 10s_2$$
$$v_1 = 12s_1 + 50s_2, \quad v_2 = 10s_1 + 40s_2.$$

What are the equilibrium bidding functions? We find α such that

$$\hat{b}_{11}(s_1) = v_{11}(s_1, \alpha) = v_{22}(s_1, \alpha).$$

Solving the right equality, we have $\alpha = s_1$ and so $\hat{b}_{11}(s_1) = 50s_1$. Similarly, we find α such that

$$\hat{b}_{12}(s_1) = v_{12}(s_1, \alpha) = v_{21}(s_1, \alpha).$$

This gives $\alpha = s_1/5$ and $\hat{b}_{12}(s_1) = 22s_1$.

For bidder 2, we find β such that

$$\hat{b}_{21}(s_2) = v_{21}(\beta, s_2) = v_{12}(\beta, s_2).$$

This gives $\beta = 5s_2$ and $\hat{b}_{21}(s_2) = 110s_2$.

Finally, we find β such that

$$\hat{b}_{22}(s_2) = v_{22}(\beta, s_2) = v_{11}(\beta, s_2).$$

This gives $\beta = s_2$ and $\hat{b}_{22}(s_2) = 50s_2$.

Let's check that this is efficient: Consider the case $s_1 = 1, s_2 = 2$. Then,

$$\hat{b}_{11}(1) = 50, \quad \hat{b}_{12}(1) = 22, \quad \hat{b}_{21}(2) = 220, \quad \hat{b}_{22}(2) = 100.$$

Thus 2 wins both units and pays $22 + 50 = 72$.

10 Collusion (Bidding Ring)

A weak cartel is one in which the members cannot do transfer payments. They blah blah.

We focus on **strong collusions**, where the members can make transfer payments. Let's return to the independent private values model and symmetric case for simplicity, with $v_i \sim f_i$ and $v_i \sim [0, 1]$. Suppose the seller uses a FPA with a reserve price ρ.

In the benchmark case (no collusion), the good is allocated to the bidder with the highest value (if larger than reserve price). The collusive case is one in which the bidders pay precisely the reserve price.

Suppose all bidders participate in the bidding ring. Clearly, the best the bidding ring can do is to win the good when $\max\{v_j\} \geq \rho$ and in that case pay the seller ρ. We'll show that this can be achieved.

The bidders do this by running a **knockout** auction among themselves: The N bidders each submit a sealed bid $\in [0, \infty)$. The highest bid b wins and earns the right tot bid ρ in the real auction. All other bid $< \rho$ in the real auction. The high bidder keeps the good and then pays each of the other bidders

$$\frac{b - \rho}{N - 1}.$$

Hence, the total payment made by the winner is

$$\rho + (N - 1) \cdot \frac{b - \rho}{N - 1} = b.$$

Now suppose $\tilde{b}(r)$ is the symmetric equilibrium bidding function for the knockout auction. Assume it is strictly increasing and $\tilde{b}(v) = 0$ if $v < \rho$ and $\tilde{b}(\rho) = \rho$. Let $u(r, v)$ be the utility when her value if v and she bids $\hat{b}(r)$ in the knockout auction. Then, we have

$$u(r, v) = F^{N-1}(r)\left(v - \tilde{b}(r)\right) + \int_r^\infty \frac{\tilde{b}(x) - \rho}{N - 1} \cdot (N - 1) f(x) F^{N-2}(x) dx.$$

We have

$$\frac{\mathrm{d}u}{\mathrm{d}r} = (N - 1) f(r) F^{N-2}(r)\left(v - \tilde{b}(r)\right) - F^{N-1}(r)\tilde{b}'(r) - (\tilde{b}(r) - \rho) F^{N-2}(r) f(r)$$

The condition $\partial_r u|_{r=v} = 0$ can be rewritten as

$$N F^{N-2}(v) f(v)(\tilde{b}(v) - \rho) + F^{N-1}(v)\tilde{b}'(v) = (N - 1) F^{N-2}(v) f(v)(v - \rho).$$

Multiplying both sides by $F(v)$, we get

$$\frac{\mathrm{d}}{\mathrm{d}v}\left[F^N(v)(\tilde{b}(v) - \varepsilon)\right] = (N - 1) F^{N-1}(v) f(v)(v - \rho), \quad v \geq \rho.$$

Integrating, we get

$$F^N(\tilde{b}(v) - \rho) - \underbrace{F^N(\varepsilon)(\tilde{b}(\rho) - \rho)}_{0} = \int_\rho^v (N - 1) F^{N-1}(x) f(x)(x - \rho) dx.$$

Thus,

$$\begin{aligned}\tilde{b}(v) - \rho &= \frac{N-1}{N} \int_{\rho}^{v} (x-\rho) \left[\frac{N F^{N-1}(x) f(x)}{F^{N}(v)} \right] \mathrm{d}x \\ &= \frac{N-1}{N} \mathbb{E}\left[\text{cartel surplus} \mid \text{highest value is } v\right].\end{aligned}$$

To verify that this is an equilibrium, look at $\partial_r u(r, v)$ and change v.